

EWATER MONITOR

API & MCP Endpoint Reference

Upstream eWater API endpoints consumed & MCP tools exposed

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Part 1 — Upstream eWater API (consumed)

These are the eWater-defined API endpoints called by this application. All requests require a Bearer token obtained from the auth endpoint.

Authentication — <https://auth.ewater.io>

POST /api/Auth/GetToken

Obtain a bearer token. Token is cached and reused; expired tokens are refreshed automatically.

Field	Type	Notes
username	string	eWater account username
password	string	eWater account password

Response: { access_token, expires_in }

Query API — <https://query.ewater.io>

GET /api/Asset/AssetHealthStatus?assetId={id}

Returns a bundled health overview for one asset.

Response: { connectivityStatus, powerStatus, flowStatus, tankHeightStatus, assetUsageStatus }

GET /api/Asset/AssetPowerStatus?assetId={id}

Latest battery voltage and trend data for an asset.

Response: { lastKnownVoltage, lastKnownVoltageReadingDt, trendDirection, todayAverage, todayHigh, todayLow, todayLowBatteryEventCount, endOfDayDeclineConsecutiveDays, endOfDayDeclineTotalVoltageDrop }

No historical voltage time-series endpoint exists. GetPowerHistoryByDateRange → 404.

GET /api/Asset/AssetFlowStatus?assetId={id}

Average flow rates for multiple periods. Response is flat (no data wrapper).

Response: { assetId, hourAverageFlowRate, todayAverageFlowRate, yesterdayAverageFlowRate, weekAverageFlowRate, monthAverageFlowRate }

Field names are *AverageFlowRate prefixed by period. NOT averageFlowRateThisHour etc. — using the wrong names silently yields null.

GET /api/Asset/AssetUsageStatus?assetId={id}

Today's usage summary. Response is flat (no data wrapper).

Response: { assetId, litresDispensedToday, lastUsageDt }

GET `/api/Entity/LitresDispensedPerDay?entityType=Asset&entityId={id}&startDt=...&endDt=...`

Per-day litres dispensed at the entity level.

Returns empty `litresDispensedPerDay` array for individual assets. May work at higher entity levels only.

State API — <https://state.ewater.io>

POST `/api/Asset/GetTankHeightHistoryByDateRange`

Time-series tank height (water + chlorine) for an asset. Auto-aggregates: hourly for ≤ 7 days, daily for longer ranges.

Field	Type	Notes
assetId	number	
startDate	string	ISO e.g. 2026-06-11T00:00:00
endDate	string	ISO e.g. 2026-06-14T23:59:59

Response: { data: [{ lowerBound, upperBound, averageWaterTankHeight, minimumWaterTankHeight, maximumWaterTankHeight, averageChlorineTankHeight, ... }] }

waterTank = vsen1, chlorineTank = vsen2. Chlorine fields all zeros if sensor not connected.

POST `/api/Asset/GetDisbursementHistoryByDateRange`

Daily water dispensed totals. Returns daily buckets; missing days produce gaps.

Field	Type	Notes
assetId	number	
startDate	string	ISO datetime
endDate	string	ISO datetime
includeTickAccumulatorDerivedDisbursement	boolean	Must be true

Response: { data: [{ lowerBound, upperBound, tickAccumulatorDerivedTotalLitres, estimateTotalLitres, totalTicks, ... }] }

Use `tickAccumulatorDerivedTotalLitres` — matches actual portal values. `estimateTotalLitres` is $\sim 30\times$ off and unreliable. `lowerBound` dates are UTC with "Z" suffix — use `.slice(0,10)` for date key.

POST `/api/Asset/GetInflowHistoryByDateRange`

Individual tap-event buckets (not daily aggregates). Used for inflow analysis.

Field	Type	Notes
assetId	number	
startDate	string	ISO datetime
endDate	string	ISO datetime

Response: { data: [{ lowerBound, upperBound, estimateTotalLitres, totalTicks, totalCredits, tickAccumulatorDerivedTotalLitres }] }

Correct formula: `totalTicks / GetTicksPerLitre = actual litres`. `estimateTotalLitres` is wrong. `tickAccumulatorDerivedTotalLitres` is null here — do NOT use for usage chart (use `GetDisbursementHistoryByDateRange` instead).

GET /api/Asset/GetTicksPerLitre?assetId={id}

LCF (ticks per litre / litres conversion factor) for an asset. Used as fallback when LitresConversion setting is absent or ≤ 0 .

Response: { ticksPerLitre, success, errorMessage }

Lives on the **state** base only. Calling on the query base → 404.

GET /api/Asset/GetIdentifiersByAssetId?assetId={id}

Returns IMEI(s) and other hardware identifiers for an asset. An asset can have multiple IMEI rows (e.g. after a device swap).

Response: { identifiers: [{ imei, ... }] }

Treat as string[]. Use the *last* non-empty IMEI (most recently registered) for commands. Rows can be blank/stale.

GET /api/Asset/GetSettingsMapForAsset?assetId={id}

Device EWC settings as key→value map. Key settings: LitresConversion (LCF, primary), Preload, FCF and other calibration values.

LCF resolution order: LitresConversion primary → GetTicksPerLitre fallback when absent/ ≤ 0 . Never fabricate a default.

POST /api/Asset/GetLogsInDateRangeByImei

Raw device packet log keyed by IMEI. Returns packets from all protocols mixed together. For assets with multiple IMEIs, fetch per-IMEI and merge.

Field	Type	Notes
imei	string	Single IMEI
startDate	string	ISO datetime
endDate	string	ISO datetime

Mixed protocols in one stream: Ewc2_5, CommandApi_1, Gadwall, CommandApi_Gadwall_1, Shengda NB-IoT. Decode against each known decoder; don't assume one protocol per asset. No upstream push/websocket — polling only (30 s interval recommended).

Command API — <https://command.ewater.io>**POST** /api/Ewc/ResetTickAccumulator

Resets the lifetime tick accumulator (meter reading) on the device to a given litre value.

Field	Type	Notes
correlationId	null	
secondaryUserId	null	
imei	string	Device IMEI (from GetIdentifiersByAssetId)
assetId	number	
litreValue	number	Target value in litres — the device does its own ticks conversion

Send litres directly — do NOT multiply by LCF. Effect is deferred until the device's next health-state packet. Requires device IMEI (not just assetId).

POST /api/Ewc/RequestSettingChange

Queues a device configuration change (e.g. LCF calibration, preload). One call per setting. Device applies on next check-in.

Field	Type	Notes
correlationId	null	
secondaryUserId	null	
assetId	number	
settingKey	string	e.g. LitresConversion, Preload
newValue	string/number	New setting value

NEVER auto-fire. Always require user confirmation — writes to real live dispensers.

Part 2 — App-Exposed MCP Tools

Endpoint: POST | GET | DELETE /api/mcp · Streamable HTTP (MCP protocol) · Auth: Authorization: Bearer <MCP_BEARER_TOKEN>

Stateless — a fresh server + transport is created per request. All tools delegate to lib/ewater-insights.ts; no eWater logic lives in the MCP layer. eWater credentials must be configured on the server before tools will respond.

Hierarchy — Country → Organisation → Water System → Asset

TOOL list_countries

All countries in the eWater hierarchy with org/water-system/asset counts. Always returned in full (hasMore=false).
No parameters.

TOOL list_organisations

Organisations, optionally filtered by country. Always returned in full.

Param	Type	Notes
countryId	number?	From list_countries
countryName	string?	Exact match

TOOL `list_water_systems`

Water systems, optionally filtered by organisation and/or country. Always returned in full.

Param	Type	Notes
organisationId	number?	
organisationName	string?	Exact match
countryId	number?	
countryName	string?	Exact match

Assets**TOOL** `list_assets`

Paginated asset list (taps/dispensers). Each item: id, name, asset type, status, GPS, water system, country.

Param	Type	Default
limit	int 1-100	50
offset	int	0
status	string?	Active · Staged · PreInstallation · Suspended
waterSystemId / waterSystemName	number? / string?	
organisationId / organisationName	number? / string?	
countryId / countryName	number? / string?	

TOOL `get_asset`

Full details for one asset: name, type, status, GPS, water system, country. Returns standard single-item envelope.

Param	Type	Notes
assetId	int	Numeric asset ID

TOOL `get_asset_history`

Time-series history: tank height (water + chlorine), daily inflow volume, battery voltage, flow rate history, dispense volumes.

Param	Type	Default
assetId	string	required
days	int 1-180	7
limit	int 1-2000	500
offset	int	0

TOOL `get_asset_ewc_settings`

EWC device configuration: LCF (litres conversion factor), FCF, preload charge, price per litre, currency/FX settings.

Param	Type	Notes
assetId	string	required

TOOL `get_asset_flow_rate`

Most-recent dispense flow rate (L/min) computed from last 24 h of device logs.

Param	Type	Notes
assetId	string	required

TOOL `get_calibration_analysis`

NRW / calibration gap analysis: LCF gap %, preload gap, plain-language summary of meter over/under-reading.

Param	Type	Default
assetId	string	required
days	int 1-180	30

Tags & Households

TOOL `list_registered_tags`

All fully-registered NFC tag IDs for a water system. totalCount = fleet size (usable without fetching all pages).

Param	Type	Default
waterSystemId	int	required
limit	int 1-500	100
offset	int	0

TOOL `get_tag` / `get_tag_info` (alias)

NFC tag lookup by 8-char hex ID. Returns primary_asset_id, household_id, credit balance, dispense/top-up counts, deleted + blacklisted flags.

Param	Type	Notes
nfcId	string	8-char hex, e.g. DBDA4ED2 — case-insensitive

Use primary_asset_id → get_asset for tap details. Use household_id → get_household for household name.

TOOL `get_tag_usage`

Dispense events for a tag at its primary asset: timestamp, litres (ticks ÷ LCF), creditConsumed, eventType.

Param	Type	Default
nfcId	string	required
days	int 1-90	30
offset	int	0
limit	int 1-200	50

TOOL `get_household` / `get_household_info` (alias)

Household name + details by UUID. Phone/address/GPS omitted (removed by eWater per data-protection policy).

Param	Type	Notes
householdId	string	UUID from get_tag

Logs

TOOL `get_asset_logs`

Decoded device packets, most-recent-first. Four decoded kinds returned inline:

kind	Description	pipeline
ewc-datalog	EWC 2.5 telemetry from device	MQTT / UDP
ewc-reply	Device reply to a platform command	MQTT / UDP
cmdapi	Outgoing command record (NOT incoming data)	CmdApi
shengda-nbiot	Shengda NB-IoT CBOR/LwM2M frame	MQTT

Param	Type	Default
assetId	string	required
days	int 1-30	7
limit	int 1-100	25
before	ISO string?	cursor for paging
protocol	string?	Ewc2_5 · CommandApi_1 · 4CCv1 · Gadwall
excludeProtocols	string[]?	Server-side exclusion list

pipeline=CmdApi entries are outgoing commands, NOT device telemetry. If all visible entries are CmdApi, the device appears offline.

Disbursements (aggregated daily totals)

TOOL **get_disbursements_by_tag_and_asset**

Daily dispense totals for one tag at one specific asset.

Param	Type	Default
nfcId	string	required
assetId	string number	required
days	int 1-365	30

*Per-day fields: estimateTotalLitres, totalTicks, totalSeconds, totalCredits (null = no-credit event), readingCount.
Also: period-level totalLitres + totalReadings.*

TOOL **get_disbursements_by_tag**

Daily dispense totals for one tag across ALL assets (multiple taps on same day summed into one bucket).

Param	Type	Default
nfcId	string	required
days	int 1-365	30

TOOL **get_disbursements_by_asset**

Daily dispense totals for one asset across ALL tags.

Param	Type	Default
assetId	string number	required
days	int 1-365	30